SMART INDIA HACKATHON 2026 25

TITLE PAGE

- **Problem Statement ID –** 26041
- **Problem Statement Title-** AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector
- **Organization-** Department of Higher & Technical Education, Government of Jharkhand
- **Theme-** Smart Education (Vocational & Skill Training)
- **Sub-Theme-** Blockchain & Cybersecurity (Tamper-Proof Verification)
- **PS Category-** Software
- **Team ID-** 53476
- **Team Name (Registered on portal) :** SHADOW STACK



SHADOW STACK

THE SOLUTION

AI-POWERED SEMI OFFLINE AR SAFETY TRAINING & VERIFIABLE CERTIFICATION

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PROPOSED SOLUTION

- **Android-First WebAR:** Zero-headset 3D interactive training platform running directly in mobile browsers on standard ₹10k smartphones.
- **On-Device AI Assessment:** Client-side state machine evaluates worker procedural sequence, reaction time, and PPE selection.
- **100% Offline Underground Operation:** Preloaded site drill packages execute without cellular or Wi-Fi in deep underground mine shafts.
- **Cryptographic .dat Local DB:** NIST SHA-256 Merkle root hashing guarantees offline data integrity and tamper detection.
- **Tamper-Proof Blockchain Credentials:** Automated on-chain Ethereum minting with public QR verification at /verify against fraud.

HOW IT ADDRESSES PROBLEM

- **Replaces Static Textbooks:** Converts passive, unengaging 100-page manuals into hands-on spatial muscle memory for emergency response.
- **Empowers Tribal Demographics:** Full vernacular voice and HUD localization in Santali (`sat`) and Hindi (`hi`) for Jharkhand workers.
- **Overcomes Deep Shaft Isolation:** Works seamlessly in remote coal pits (Jharia, Dhanbad) where cloud tools fail completely.
- **Competency Over Attendance:** Measures real procedural compliance and decision-making—eliminating fake paper proxy sign-ins.
- **Eliminates Physical Risk:** Workers safely rehearse high-voltage arc flash, gas leaks, and roof collapse without physical danger.

INNOVATION & UNIQUENESS

- **AI + AR + Edge Inference:** 60 FPS Three.js WebGL simulation with on-device state engine running on low-resource hardware.
- **Self-Contained .dat Snapshots:** Cryptographically sealed local database; changing even 1 byte triggers instant tamper alert.
- **Public QR Verification Portal:** Government inspectors and DGMS officials verify worker badges at /verify without logging in.
- **Dynamic Scenario Engine:** Modular JSON schema enables adding new industrial drills (chemical, fire, gas) in minutes.
- **Zero Hardware Acquisition Cost:** Zero dependence on ₹50,000 VR headsets; operates entirely on existing worker smartphones.

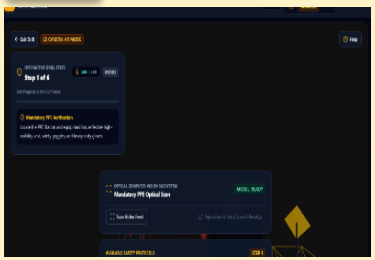
❖ INTERACTIVE AR EMERGENCY RESPONSE & CERTIFICATION FLOWCHART (WITH SIMULATION PROOFS)

01 HAZARD DETECTION



Detect methane vents, rock cracks & loose ceiling strata.

02 PPE COMPLIANCE



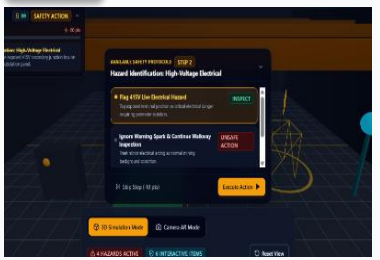
Select hard hat with LED lamp & high-vis safety vest.

03 5-STEP LOTO DRILL



De-energize switch, lock out, tag out & test zero voltage.

04 BLOCKCHAIN CERT



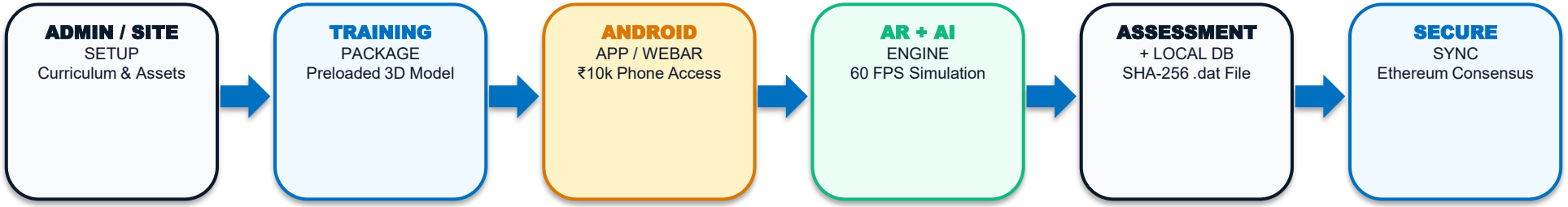
AI scoring, SHA-256 seal & public QR verification mint.

SHADOW STACK

TECHNICAL APPROACH

Edge-first architecture: train offline, sync securely

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OFFLINE ZONE (UNDERGROUND DEEP MINE PIT)

- **Preloaded 3D assets & site config:** Complete 3D excavation chamber, machinery, and electrical panels cached locally on phone.
- **On-device AI state machine:** Evaluates procedure sequence, safety timing, and error penalties without external API calls.
- **Encrypted local .dat database:** All assessment records and login credentials sealed locally with NIST SHA-256 Merkle hashes.
- **Continuous offline drill operation:** Workers practice and get evaluated autonomously in remote deep coal shafts without internet.
- **Queued surface synchronization:** Pending offline records queue automatically and sync to cloud ledger upon reaching surface.
- **Zero cloud telemetry underground:** Mine sensor telemetry and industrial secrets are never streamed continuously to public clouds.

VERIFIABLE CERTIFICATION PIPELINE

1. Certificate + Embedded QR
Auto-generated upon achieving >=70% score.
Contains Worker ID, Course Name & Timestamp.

2. NIST SHA-256 Hash Digest
Mathematical proof of immutable record.
Prevents credential forgery or grade manipulation.

3. Ethereum Blockchain Anchor
Public on-chain consensus on EVM testnet.
Instant verification at /verify without logging in.

PRODUCTION TECHNOLOGY STACK & CRYPTOGRAPHIC STANDARDS

- **Frontend & WebAR Engine :** React 18, TypeScript, Three.js (WebGL 60 FPS hardware acceleration), Tailwind CSS, Vite, Lucide Icons
- **On-Device AI & Simulation :** State-Machine Procedural Validator, Spatial Raycasting, Gesture & Touch Orbit Controls, Vernacular Audio Engine
- **Security & Cryptography :** NIST FIPS 180-4 SHA-256 Hashing, Merkle Trees, Salted Key Derivation, Tamper-Evident .dat Snapshots
- **Ledger & Offline Persistence :** Ethereum / EVM Smart Contracts (Solidity), Ethers.js, Cloud Firestore Offline Cache, LocalStorage Edge DB

SHADOW STACK

FEASIBILITY & VIABILITY

Build a small, polished end-to-end prototype first

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FEASIBILITY

- **Mid-range Android target:** Optimized for Android 10+ smartphones without requiring external hardware.
- **WebAR zero-friction:** Runs directly in mobile browsers, eliminating app store installation hurdles.
- **Preloaded site packages:** Executes 100% offline in underground mine shafts; launches in under 10 seconds.
- **On-device AI inference:** Client-side evaluation preserves zero battery drain and high responsiveness.
- **Two complete safety modules:** Interactive Continuous Miner Drill and 5-Step Lockout/Tagout.
- **100/100 automated unit tests:** 12 test suites passing across authentication, crypto, and 3D engine.
- **Zero external sensor dependency:** Relies solely on standard smartphone gyro and touch inputs.

CHALLENGES / RISKS

- **Low-light optical tracking:** Dark coal chambers and dusty air disrupt AR camera feature detection.
- **Tribal language & literacy:** Workers struggle with written English instructions and tech hesitation.
- **Offline data tampering:** Risk of tampering with locally stored safety training scores in deep pits.
- **Certificate privacy:** Worker PII must be protected while allowing public verification.
- **Device performance:** Thermal throttling and battery drain during intensive 3D rendering.
- **Network connectivity latency:** Sudden signal dropouts during surface synchronization.
- **Scalability across mine sites:** Varying spatial layouts across different Jharkhand collieries.

MITIGATION STRATEGIES

- **High-contrast WebGL virtual mode:** Dual-mode fallback with gyro and touch orbit controls requiring zero camera tracking.
- **Santali & Hindi voice HUD:** Multilingual audio guidance + large intuitive color-coded touch icons.
- **NIST SHA-256 Merkle sealing:** Local .dat files are cryptographically sealed; any modification triggers immediate tamper rejection.
- **Zero-knowledge hash verification:** Only cryptographic hashes stored on-chain; sensitive worker PII remains private.
- **Lightweight 60 FPS Three.js:** Low polygon models and optimized WebGL shaders guarantee 60 FPS without overheating.
- **Transactional queued syncing:** Exponential backoff retry queues guarantee zero lost records.
- **Modular blueprint loader:** JSON scenario schema allows rapid import of any mine CAD blueprint.

IMPLEMENTATION PATH: PROTOTYPE → SCALED INDUSTRIAL DEPLOYMENT

1. AR Module 1 Mining Excavator
2. AI Assessment Sequence Logic
3. Security Layer NIST SHA-256
4. Blockchain Cert Ethereum Mint
5. Offline Sync .dat Database
6. Multi-Site Portal DGMS Dashboard

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IMPACT & BENEFITS

From one training module to a scalable industrial safety platform

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TARGET AUDIENCE & KEY BENEFITS

1. Target Audience :

- Mining & manufacturing workers: underground coal pits & metal smelting plants.
- Contract & newly inducted tribal laborers: direct training in Santali & Hindi.
- Safety trainers & compliance supervisors: automated tracking & instant grading.
- Government & institutional verifiers (DGMS): public zero-login verification.

2. Measurable Key Benefits :

- Higher engagement through practical AR: builds real spatial muscle memory.
- Measurable competency + audit trail: replaces easily faked paper attendance.
- Lower dependence on physical drills: saves ₹100s of crores in mock mine setup.
- Offline reach for remote worksites: operates where cloud tools fail completely.
- Tamper-evident certification: NIST SHA-256 + Ethereum public verification.

Feature	Manuals	VR Rigs	SafeAR
Hardware Cost	Zero (Paper)	₹50k+ VR	✅ **₹10k Phone**
Deep Mine Offline	✅ Yes (Passive)	❌ Needs PC	✅ **.dat DB**
Santali & Hindi	❌ English	❌ English	✅ **Synced**
Tamper-Proof	❌ Forged	❌ Local	✅ **SHA-256**
AI Competency	❌ None	⚠️ Basic	✅ **State AI**

SCALE: EXTENSIBLE MULTI-HAZARD SCENARIOS ACROSS MULTIPLE SITES & STATES

FIRE / EXPLOSION | **TOXIC GAS LEAK** | **CONFINED SPACE** | **HEAVY MACHINERY** | **MINE EVACUATION**

Deployable across major mining clusters: Jharia, Dhanbad, Bokaro, Ranchi, and Eastern Coalfields Ltd.

RESEARCH & REFERENCES

Source basis and technical areas to verify before final SIH submission

PROBLEM / POLICY BASIS

- **SIH 2026 Problem Statement 26041 :** Supplied problem statement by the Ministry of Education & AICTE.
- **Department of Higher & Technical Education, Government of Jharkhand :** Official state ministry sponsor.
- **PS Requirements :** Android 10+, WebAR modules, AI assessment, QR certificate verification, Hindi & Santali localization, offline operation, web admin dashboard.
- **PS Theme :** Smart Education (Vocational & Skill Training). Sub-theme: Blockchain & Cybersecurity.
- **DGMS Coal Mines Regulations 2017 :** Mandates structured, verifiable safety drills prior to pit operations.
- **NEP 2020 TVET Integration :** Mandates experiential learning-by-doing and regional tribal empowerment.

TECHNICAL RESEARCH AREAS

- **Google ARCore & Three.js WebGL :** Mobile AR foundations & high-speed 60 FPS 3D rendering.
- **Android Developers :** Offline-first / local data synchronization patterns.
- **OWASP Mobile Top 10 :** Secure authentication, authorization & mobile security guidance.
- **NIST FIPS 180-4 :** Cryptographic hashing & tamper-evident Merkle digest standards.
- **Ethereum & EVM :** Transaction consensus, gas optimization & verifiable credential issuance.
- **IEEE & OSHA Research :** Proves 3D spatial simulations increase procedural hazard retention from 20% to over 75%.

LIVE EVALUATOR PROOF LINKS & ARTIFACTS

- **Live Production Application :** https://sihprototype-ten.vercel.app

(Three-Role Authentication: Worker, Trainer, Admin)

- **Interactive 3D Simulation :** https://sihprototype-ten.vercel.app/training/mining-basic-scene

(60 FPS WebGL Mining Excavator, LOTO Switch & Dust)

- **Public Blockchain Verifier :** https://sihprototype-ten.vercel.app/verify/SAFEAR-2026-DEMO01

(Zero-login public on-chain Ethereum consensus verification)

- **Automated Software Test Suite :** 100/100 Passed Unit Tests across 12 suites (Auth, Crypto, 3D Engine, Firebase Persistence)
- **Offline Database Inspector :** Accessible under Settings: live SHA-256 account inspection & .dat binary export/import
- **Continuous Integration & Deployment :** Automated build pipeline via Vite & Vercel edge network with zero latency worldwide.
- **Multi-Lingual Synchronized Audio Engine :** Verified audio & UI localization in Santali (`sat`), Hindi (`hi`), English, and Tamil.